

CACS101: Computer Fundamentals and Applications

Unit 5 – Data Communication and Computer Network

PRAJWAL BISTA

Data Communication

- Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.
- For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).
- The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

Data Communication

- **Delivery:** The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.
- **Accuracy:** The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable
- **Timeliness:** The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.
- **Jitter:** Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets

Components of data communication system

- **Message:** The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
- **Sender:** The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
- **Receiver:** The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
- **Transmission medium:** The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
- **Protocol:** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices.

Modes of Communication

- The direction in which data can be transmitted between any two linked devices is of three types:
 1. Simplex,
 2. Half-duplex, and
 3. Full- duplex, or duplex.

Contd.

- In **simplex mode** the data transmits in one direction only, from one system to another system. The sender device that sends data can only send data and cannot receive it. On the other hand, the receiver device can only receive the data and cannot send it. Television is an example of simplex mode transmission as the broadcast sends signals to our TV but never receives signals back from our TV. This is a unidirectional transmission.

Contd.

- In **half-duplex** mode the communication between sender and receiver occurs in both directions, but only one at a time. The sender and receiver can both send and receive the information, but only one is allowed to send at any given time. For example, in walkie-talkies, the speakers at both ends can speak, but they must speak one by one. They cannot speak simultaneously.

Contd.

- In **full duplex** transmission mode, the communication between sender and receiver can occur simultaneously. The sender and receiver can both transmit and receive at the same time. For example, in a telephone conversation, two people communicate, and both are free to speak and listen at the same time.

Computer Network

- A computer network is a group of computer systems and other computing hardware devices that are linked together through communication channels to facilitate communication and resource-sharing among a wide range of users. Networks are commonly categorized based on their characteristics.
- Networks are used to:
 - Facilitate communication via email, video conferencing, instant messaging, etc.
 - Enable multiple users to share a single hardware device like a printer or scanner
 - Enable file sharing across the network
 - Allow for the sharing of software or operating programs on remote systems
 - Make information easier to access and maintain among network users

Network Criteria

- **Performance:** Performance can be measured in many ways, including transit time and response time. Transit time is the amount of time required for a message to travel from one device to another. Response time is the elapsed time between an inquiry and a response. The performance of a network depends on several factors, including the number of users, the type of transmission medium, the capabilities of the connected hardware, and the efficiency of the software.
- **Reliability:** Network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.
- **Security:** Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

Types of Computer Network

- A computer network is a group of computers linked to each other through a transmission medium that enables the computer to communicate with another computer and share their resources, data, and applications.
- There are mainly three types of computer networks based on their **size**:
 1. Local Area Network (LAN)
 2. Metropolitan Area Network (MAN)
 3. Wide area network (WAN)

Local Area Network (LAN)

- A local area network (LAN) is a network that connects computers and devices in a limited geographical area such as home, school, computer laboratory, office building, or closely positioned group of buildings.
- LAN's enable the sharing of resources such as files or hardware devices that may be needed by multiple users.
- It is limited in size, typically spanning a few hundred meters, and no more than a mile.
- It is fast, with speeds from 10 Mbps to 100 Mbps.
- LAN's can be either wired or wireless. Twisted pair, coax or fiber optic cable can be used in wired LAN's.
- Nodes in a LAN are linked together with a certain topology. These topologies include Bus ,Ring & Star

Metropolitan Area Network (MAN)

- A metropolitan area network (MAN) is a large computer network that usually spans a city or a large campus.
- A MAN is optimized for a larger geographical area than a LAN, ranging from several blocks of buildings to entire cities.
- A MAN might be owned and operated by a single organization, but it usually will be used by many individuals and organizations.
- A MAN often acts as a high-speed network to allow sharing of regional resources.
- Examples of MAN: Telephone company network that provides a high-speed DSL to customers and cable TV network.

Wide Area Network (WAN)

- WAN covers a large geographic area such as country, continent or even whole of the world.
- A WAN is two or more LANs connected. The LANs can be many miles apart.
- To cover great distances, WANs may transmit data over leased high-speed phone lines or wireless links such as satellites.
- Multiple LANs can be connected using devices such as bridges, routers, or gateways, which enable them to share data.
- The world's most popular WAN is the Internet.

LAN Topologies

- Network topology refers to the layout of connected devices, i.e., how the computers, cables, and other components within a data communications network are interconnected, both physically and logically.
- The physical topology describes how the network is laid out, and the logical topology describes how the data flow through the network.
- The topologies commonly used are Bus topology, Star topology, and Ring topology. Other topologies are Tree topology, Mesh topology and Hybrid topology.

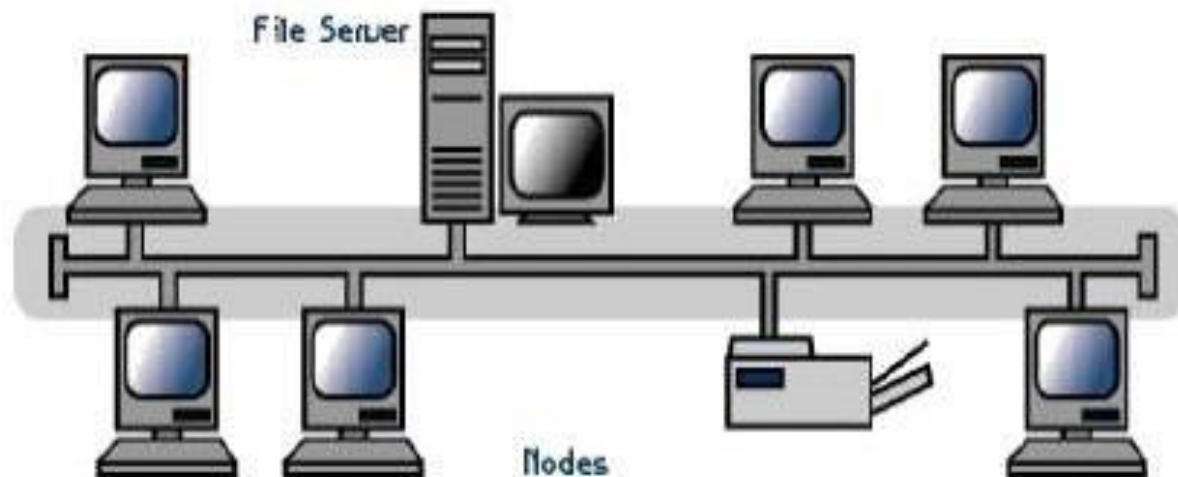
Bus Topology

- Each machine is connected to a single cable.
- Each computer or server is connected to the single bus cable through connector.
- A signal from the source travels in both directions to all machines connected on the bus cable until it finds the address on the network that is the intended recipient.
- If the machine address does not match the intended address for the data, the machine ignores the data.
- Alternatively, if the data does match the machine address, the data is accepted.

Bus Topology

- Advantages:
 1. If any node or cable (except backbone) fails, it does not affect the whole network.
 2. They are relatively cheap and easy to install
 3. Don't require much cabling
 4. Easy to add and remove any node or cable except the backbone cable
- Disadvantages:
 1. If the backbone cable fails, the entire network goes down
 2. Since all the data are transmitted through the backbone cable, data traffic is high
 3. Bus networks work best with a limited number of devices.

Bus Topology



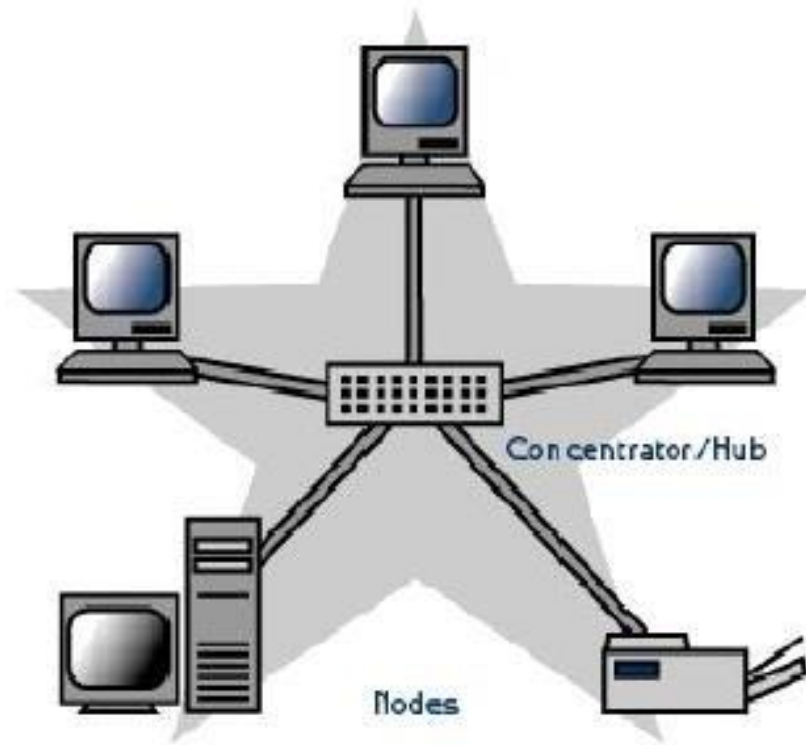
Star Topology

- Each machine is connected to a central hub or switch.
- It allows each machine on the network to have a point-to-point connection to the central hub.
- All the traffic which transverses the network passes through the central hub.
- The hub acts as a signal booster or repeater which in turn allows the signal to travel greater distances.
- Most widely implemented
- Hub is the single point of failure

Star Topology

- Advantages:
 1. A failure in any node or cable will only take down one computer's network access and not the entire LAN.
 2. Easy to diagnose the fault
 3. Easy to add and remove a node or cable
- Disadvantages:
 1. Compared to the bus topology, a star network generally requires more cable
 2. If the hub fails, the entire network also fails

Star Topology



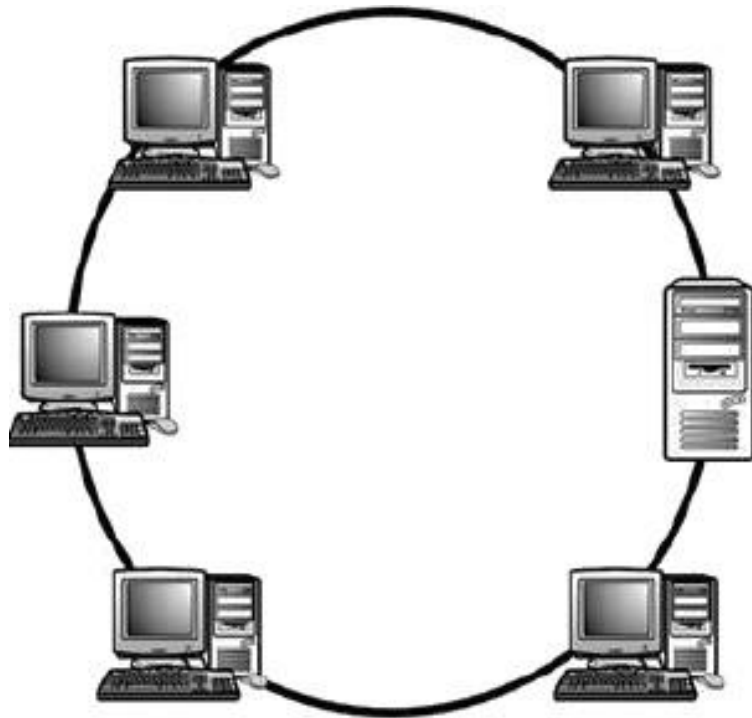
Ring Topology

- Each computer is connected to the network in a closed loop or ring
- Each machine or computer has a unique address that is used for identification purposes
- The signal passes through each machine or computer connected to the ring in one direction
- Ring topologies typically utilize a token passing scheme, used to control access to the network
- By utilizing this scheme, only one machine can transmit on the network at a time.

Ring Topology

- Advantages:
 1. Short cables are required compared to other topologies
 2. It works where there is no central site computer system
 3. It is not dependent in a single cable or node
- Disadvantages:
 1. A failure in any cable or device breaks the loop and can take down the entire network.
 2. It is rarely used in modern LAN
 3. It is difficult to find out the error
 4. Adding and removing any node or cable is difficult

Ring Topology



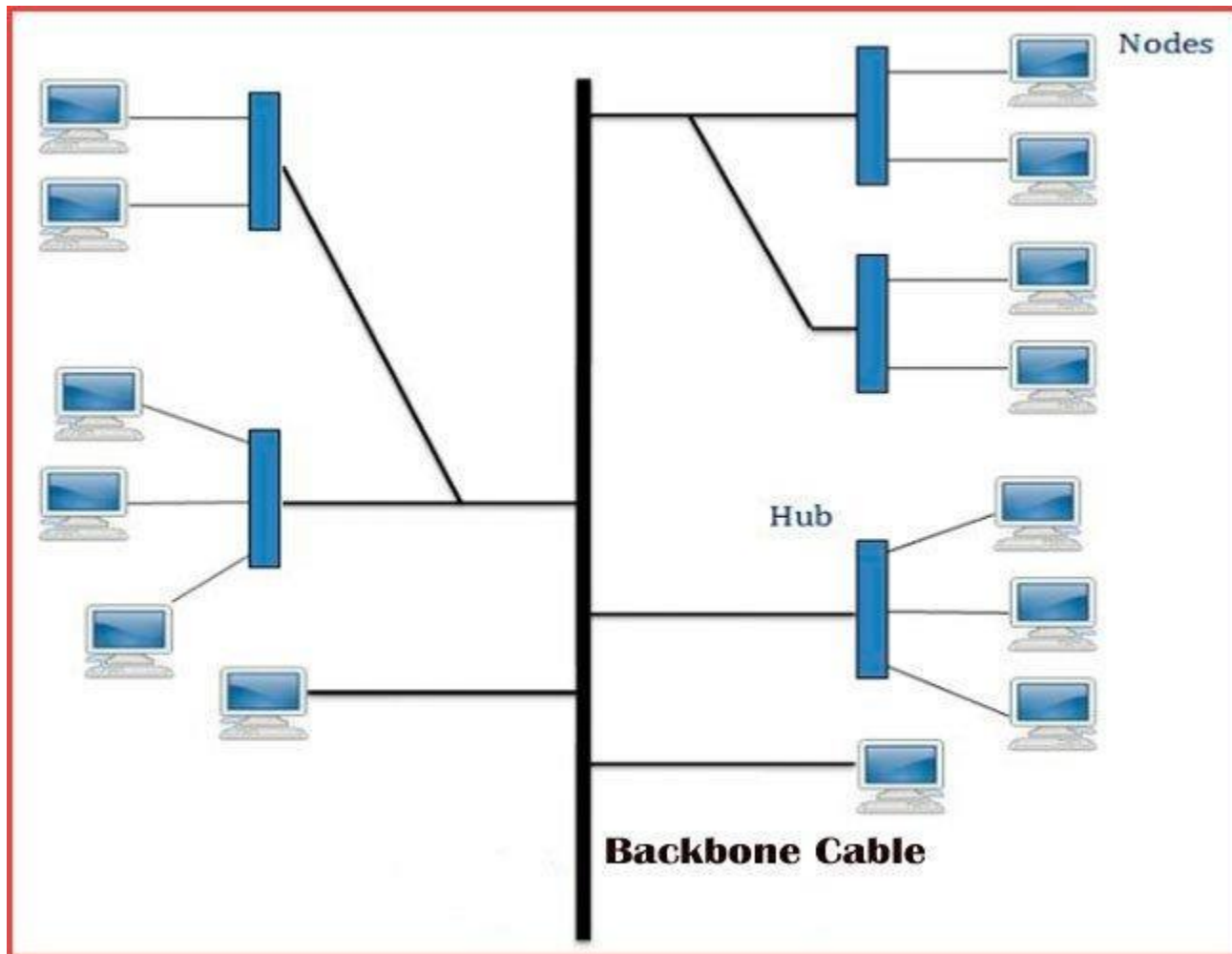
Tree Topology

- Tree topology uses bus and star topology together to create a network. This topology combines many star topologies using the main cable.
- The whole network is divided into many chunks that can be easily managed and maintained.
- In this topology, all devices are connected to the main cable in the form of star topologies.
- Tree topology is similar to bus topology because all the devices are connected to the main cable. So, the main cable acts as a backbone of the network.

Tree Topology

- Advantages:
 1. Expansion of nodes is possible and easy.
 2. If any devices damaged, other hierarchical networks are not affected.
 3. Other segments are not affected if any node or any segment gets damage.
 4. Easy to manage and maintain.
- Disadvantages:
 1. If the main backbone line breaks, then the entire network shut down.
 2. Huge amount of cable is needed.
 3. Maintenance is difficult if more devices are added.
 4. If the hub or concentrator fails, attached nodes are also disabled.

Tree Topology



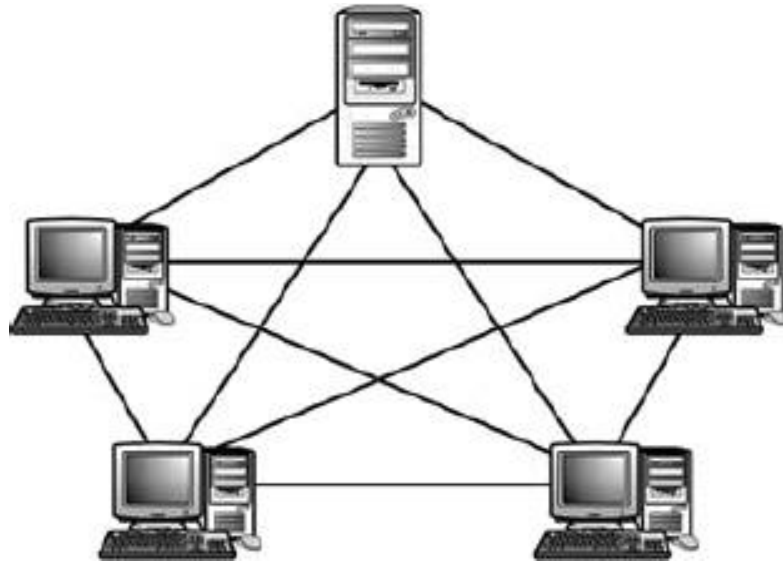
Mesh Topology

- In mesh topology, each computer on the network connects to every other, creating a point-to-point connection between every device on the network.
- Messages sent on a mesh network can take any of several possible paths from source to destination.
- It does not contain the switch, hub or any central computer which acts as a central point of communication.

Mesh Topology

- Advantages:
 1. Failure in any one wire or node does not fail the entire network
 2. Manages high amounts of traffic, because multiple devices can transmit data simultaneously.
 3. Faster communication because of point-to-point connection
- Disadvantages:
 1. Most complicated network topology
 2. Expensive in comparison to other topologies.
 3. Building and maintaining the topology is difficult and time consuming.

Mesh Topology



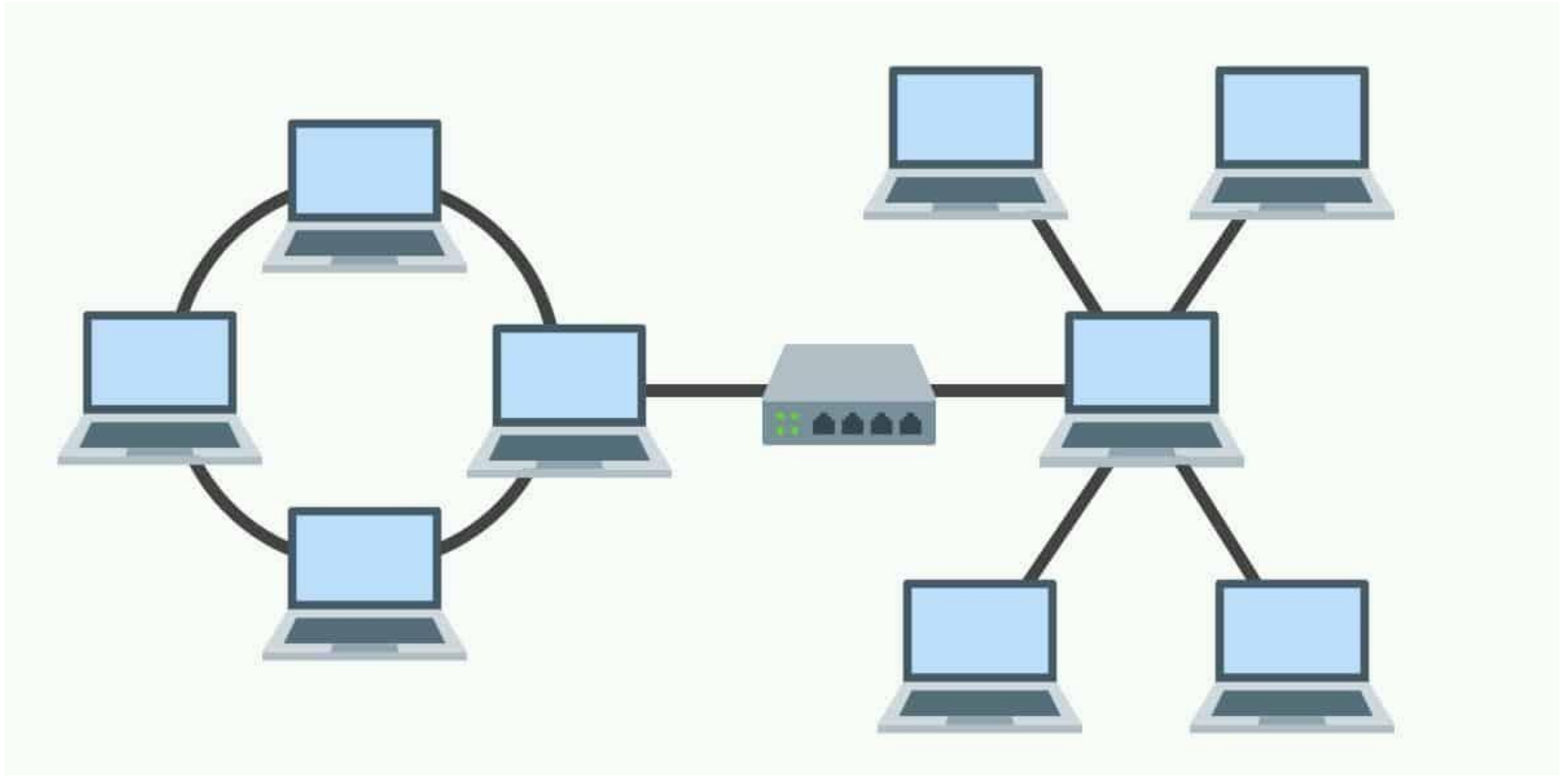
Hybrid Topology

- This topology (sometimes called mixed topology) is simply combining various topologies to form a larger, more complex topology.
- Main aim is being able to share the advantages of different topologies.

Hybrid Topology

- Advantages:
 1. Easy network growth
 2. Manages troubleshooting
- Disadvantages:
 1. Expensive network management.
 2. To manage the topology become challenging, as the different topologies are linked in the hybrid topology.

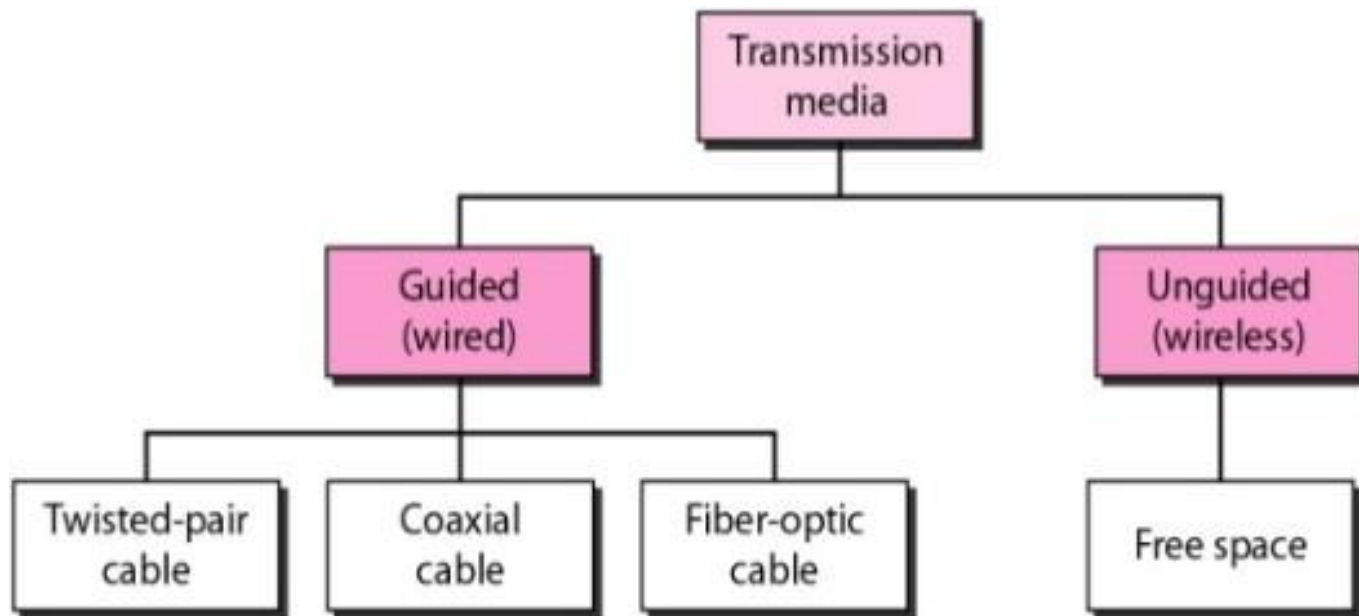
Hybrid Topology



Transmission Media

- A transmission medium can be broadly defined as anything that can carry information from a source to a destination.
- The transmission medium is usually free space, metallic cable, or fiber-optic cable.
- Transmission media can be generally categorized as either unguided or guided.

Transmission Media



Guided Media

- In the guided media, the data signals are sent along a specific path, through a wire or a cable.
- Copper wire and optical fibers are the most used guided media that transmits data as electric signals.
- Copper wires offer low resistance to current signals, facilitating signals to travel longer distances.
- Types of guided media:
 - Twisted Pair
 - Coaxial Pair
 - Optical fiber

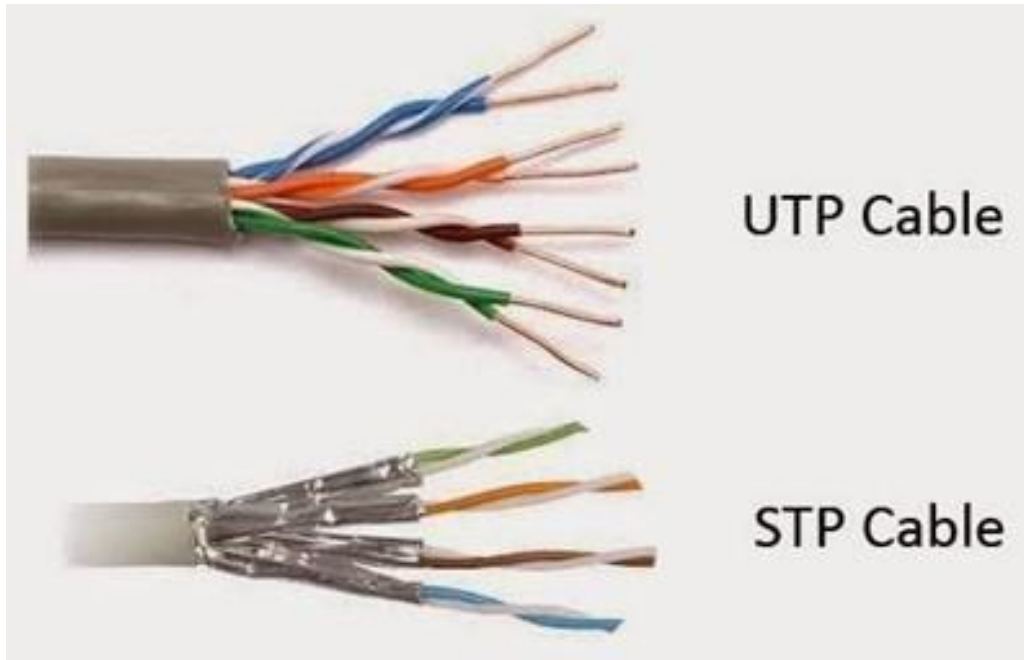
Twisted Pair Cable

- A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together.
- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.
- Twisting tends to decrease the crosstalk
- Crosstalk is the interference due to the magnetic field of 2 wires nearby
- Used to transmit both analog and digital transmission
- The two basic types of twisted-pair transmission lines specified are unshielded twisted pair (UTP) and shielded twisted pair (STP).

Twisted Pair Cable

- **STP** cable has an extra layer of metal foil between the twisted pair of copper wires and the outer covering. The metal foil covering provides additional protection from external disturbances. STP cables are costlier when compared to UTP but has the advantage of being capable of supporting higher transmission rates across longer distances.
- **UTP** is the most used medium for transmission over short distances up to 100m. Out of the four pairs of wires in a UTP cable, only two pairs are used for communication.

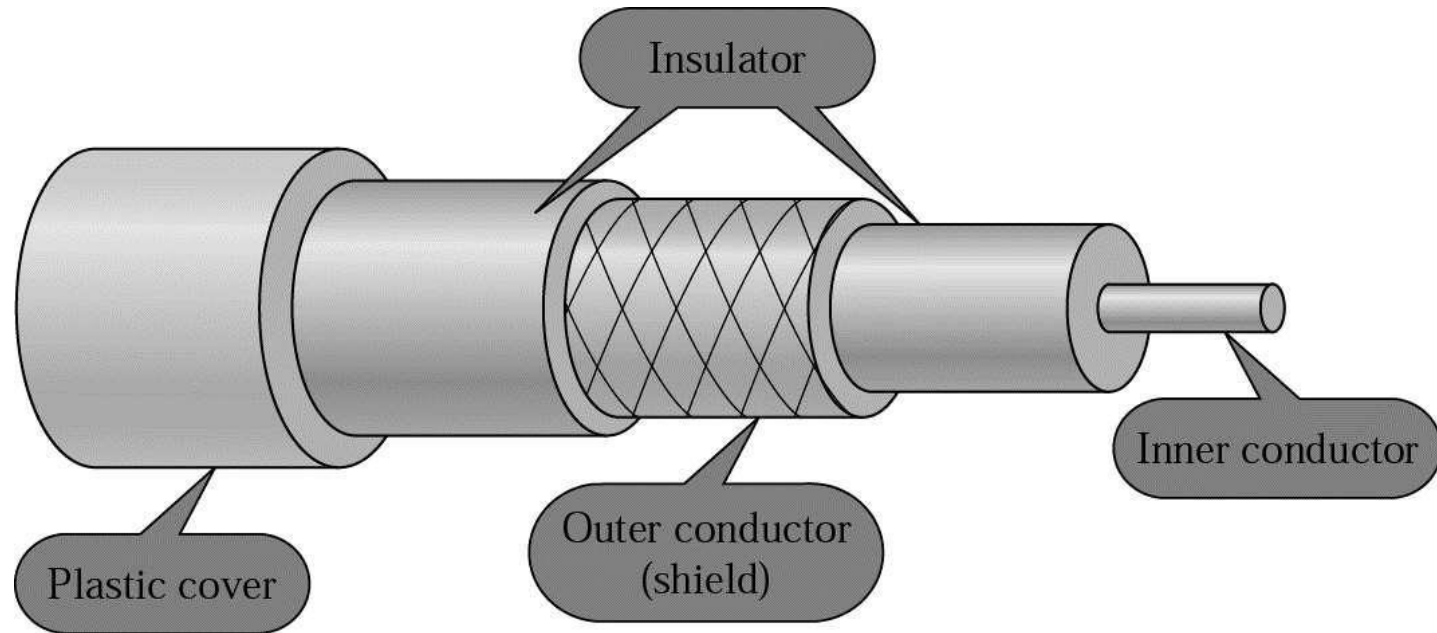
Twisted Pair Cable



Coaxial Cable

- Coaxial cable has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two.
- The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit.
- This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover.
- Coaxial cable can be used over longer distances and support more stations on a shared line than twisted pair
- Coaxial cable is a versatile transmission medium, used in a wide variety of applications, including Television distribution - aerial to TV systems

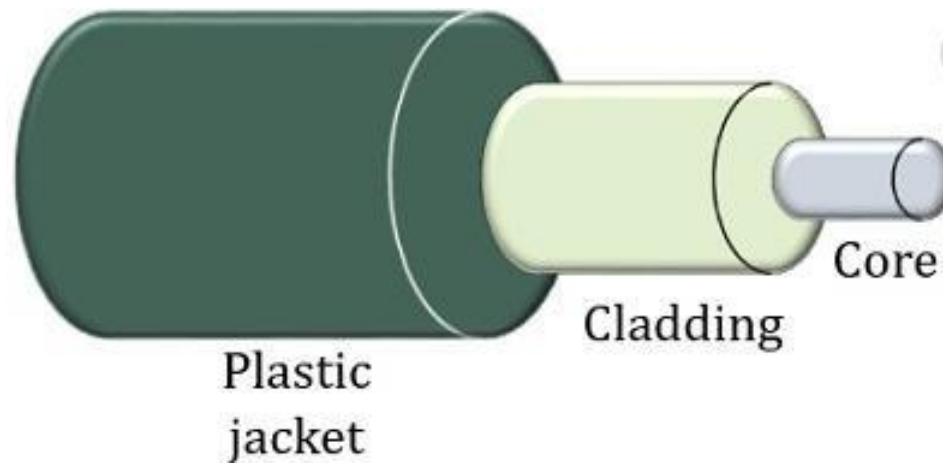
Coaxial Cable



Fiber Optic Cable

- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light. An optical fiber cable consists of
 1. **Core** - optical fiber conductor (glass) that transmits light,
 2. **Cladding** - an optical material that surrounds the core to prevent any light from escaping the core, and
 3. **Jacket** - outer covering made of plastic to protect the fiber from damage.
- Optical fibers are being used for transmission of information over large distances more cost effectively than the copper wire connection. Communication systems are now unthinkable without fiber optics.

Fiber Optic Cable



Unguided Media

- Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.
- Radio, microwave and satellite transmissions fall into this category.

Radio Transmission

- Electromagnetic waves ranging in frequencies between 3 kHz and 300 GHz are called radio waves.
- Omnidirectional, propagate in all directions. So, sending and receiving antennas do not have to be aligned and the signals can be received by all the radios within the broadcast area.
- Radio waves propagate in the sky mode, can travel long distance making it a good candidate for long distance broadcasting.
- Easy to generate
- Applications: AM & FM radio

Microwave Transmission

- The microwave is another type of electromagnetic waves that have a frequency range of 300 MHz to 300 GHz.
- Microwave transmission can be aimed at a single direction, instead of broadcasting in all directions (like in radio waves). Microwaves can carry more information than radio waves but cannot penetrate metals.
- Microwaves are used where there is a clear path between the transmitter and the receiver.
- Microwave transmission has the advantage of not requiring access to all contiguous land along the path of the system, since it does not need cables.
- It is used in satellite communication, navigation, radar, remote sensing and other short distance communication systems.
- Disadvantages
 - Needs expensive towers and repeaters, and
 - They are subject to interference from passing airplanes and rain.

Infrared waves

- An infrared transmission is a wireless technology used for communication over short ranges.
- The frequency of the infrared is in the range from 300 GHz to 400 THz.
- It is used for short-range communication such as data transfer between two cell phones, TV remote operation, wireless speakers, data transfer between a computer and cell phone resides in the same closed area.

Network Devices

- Network Devices are the devices that connects two or more LANS, or segments with each other or with the Internet.
- These connecting devices can operate in different layers of the Internet model. Some of them are:
 - Network Interface Card (NIC)
 - Repeater
 - Bridge
 - Hub
 - Switch
 - Router
 - Gateway

Network Interface Card (NIC)

- A Network Interface Card (NIC) is a hardware device through which the computer connects to a network.
- It works at both the data link layer and physical layer of the OSI reference model.
- At the data link layer, NIC converts the data packets into data frames, adds the Media Access address (MAC address) to data frames.
- At the physical layer, it converts the data into signals and transmits it across the communication medium.

Repeater

- Repeaters are used to extend LAN. It has only two ports and can connect only two segments of the network.
- It retimes and regenerates the signals to proper amplitudes and sends them to the other segments.
- Signals that carry information within a network can travel a fixed distance before attenuation endangers the integrity of the data.
- A repeater receives a signal and before it becomes too weak or corrupted, regenerates the original bit pattern. The repeater then sends the refreshed signal.

Repeater

- It can extend the physical length of a LAN.
- Repeaters require a small amount of time to regenerate the signal. This can cause a propagation delay which can affect network communication when there are several repeaters in a row.
- Many network architectures limit the number of repeaters that can be used in a row.
- Repeaters work only at the physical layer of the OSI network model.

Bridge

- Bridge is used to connect two LAN segments like a repeater; it forwards complete and correct frames to the other segment.
- It works in both the physical and the data link layer.
- As a physical layer device, it regenerates the signal it receives. As a data link layer device, the bridge can check the physical (MAC) address contained in the frame.
- A bridge has a filtering capabilities i.e. it can check the destination address of a frame and decide if the frame should be forwarded or dropped.

Bridge

- A bridge reads the outermost section of data on the data packet, to tell where the message is going.
- It reduces the traffic on other network segments, since it does not send all packets.
- Bridges can be programmed to reject packets from particular networks.
- Bridges forward all broadcast messages.
- Only a special bridge called a translation bridge will allow two networks of different architectures to be connected. Bridges do not normally allow connection of networks with different architectures.

Hub

- It is a device that centrally connects devices in a computer network.
- There are two types of a hub i.e. **active hub** which is a multiport repeater that is used to create connections between stations in a physical star topology; and **passive hub** which is just a connector that connects the wires coming from different branches.
- Every computer is directly connected with the hub. When data packets arrives at hub, it broadcast them to all the LAN cards in a network and the destined recipient picks them and all other computers discard the data packets.

Hub

- Hub has four, eight, sixteen and more ports and one port is known as uplink port, which is used to connect with the next hub.
- Hubs work at the physical layer of the OSI (Open System Interconnection) model. The device is a form of multiport repeater.
- Repeater hubs also participate in collision detection, forwarding a jam signal to all ports if it detects a collision.

Switch

- Like hub, switch also connects multiple computers in a network or different segments of the same network.
- Switches work at the Data Link Layer of the OSI reference model. Hence, switches consider data as frames and not as signals. Unlike the hubs, a switch does not broadcast the data to all the computers; it sends the data packets only to the destined computer.
- A switch receives a signal as a data frame from a source computer on a port, checks the MAC address of the frame, forwards the frame to the port connected to the destination computer having the same MAC addresses, reconverts the frame back into signal and sends to the destination computer.

Router

- A router is a communication device that is used to connect two logically and physically different networks, two LANs, two WANs and a LAN with WAN.
- The main function of the router is sorting and the distribution (i.e. routing) of the data packets to their destinations based on their IP addresses.
- Routers provides the connectivity between the enterprise businesses, ISPs and in the internet infrastructure, router is a main device.

Router

- Every router has routing software, which is known as IOS.
- Router operates at the network layer of the OSI model.
- Router does not broadcast the data packets. Routers connect two or more logical subnets, each having a different network address.
- A router determines which way is the shortest or fastest in a network, and routes packets accordingly based on the IP addresses.

Gateway

- Gateway is a generic term used to represent devices that connect two dissimilar networks.
- A gateway at the transport layer converts protocols among communications networks. It can accept a packet formatted for one protocol and convert it to a packet formatted for another protocol.
- An application gateway can translate messages from one format to the other. A gateway can be implemented in hardware, software, or in both hardware and software. Generally, gateway is implemented by software installed within a router.

Communication Protocol

- A protocol is a set of rules that make communication on a network more efficient.
- A data communications protocol is a set of rules or an agreement that determines the format and transmission of data.
- The network communication protocol is organized as a stack of layers with one layer built upon the other. Each layer has a specific function and interacts with the layers above and below it.
- The outgoing data from a computer connected to the network passes down through each layer and the incoming data passes up through each layer.
- The corresponding layers on the different machines are called peers. The peers interact with each other using the protocol.

OSI Reference Model

- The International Standards Organization (ISO) has developed a seven-layer reference model for data networks, known as Open System Interconnection (OSI) model.
- The OSI model specifies the functions of each layer. It does not specify how the protocol needs to be implemented.
- It is independent of the underlying architecture of the system and is thus an open system.

Seven layers of the OSI model



Physical Layer

- This layer specifies the basic network hardware.
- Some of the characteristics defined in the specification are - interface between transmission media and device, encoding of bits, bit rate, error detection parameters, network topology, and the mode of transmission (duplex, half-duplex or simplex).
- Layer 1 is anything that carries 1's and 0's between two nodes.

Data Link Layer

- This layer specifies the functions required for node-to-node transmission without errors.
- It specifies the organization of data into frames, error detection in frames during transmission, and how to transmit frames over a network.
- Data Link layer is to deliver packets from one NIC to another.
- Layer 2 uses MAC addresses and is responsible for packet delivery from hop to hop.

Network Layer

- The network layer specifies the assignment of addresses (address structure, length of address etc.) to the packets and forwarding of packets to the destination i.e. routing.
- Layer 3 uses IP addresses and is responsible for packet delivery from end to end.

Transport Layer

- It specifies the details to handle reliable transfer of data.
- It handles end-to-end error control and flow control, breaking up data into frames and reassembling the frames.
- Layer 4 is responsible for service to service delivery.

Session Layer

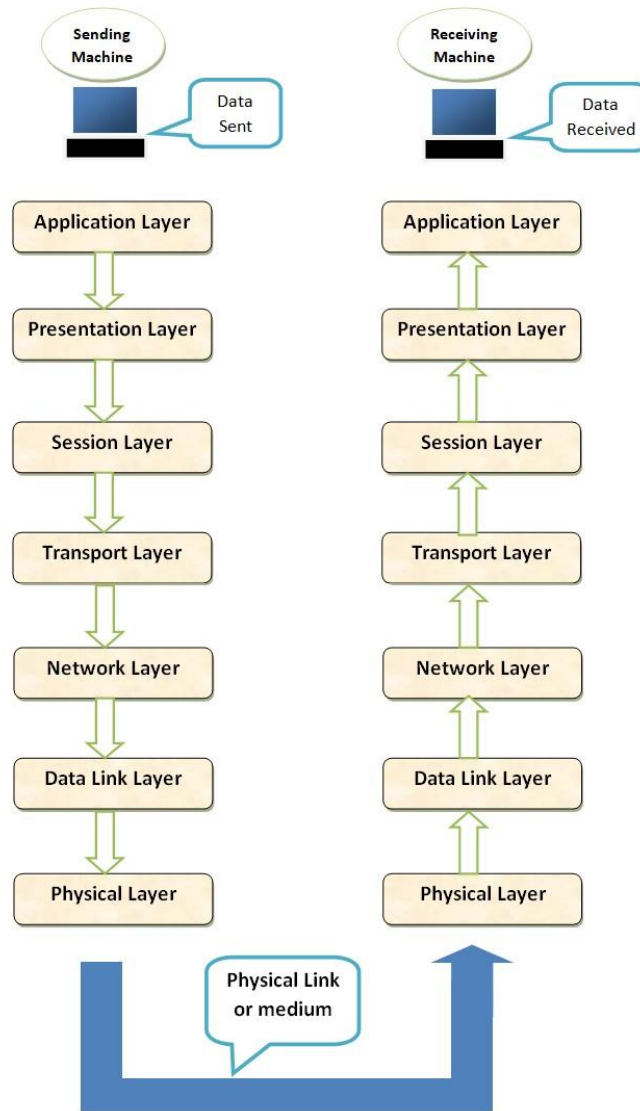
- The session layer maintains a session between the communicating devices.
- It includes specifications for password and authentication, and maintaining synchronization between the sender and the receiver.

Presentation Layer

- This layer specifies the presentation and representation of data.
- Its functions include translation of the representation of the data into an identifiable format at the receiver end, encryption, and decryption of data etc.

Application Layer

- This layer specifies how an application uses a network.
- It deals with the services attached to the data.
- It contains the protocols used by users like HTTP, protocol for file transfer and electronic mail.



- Each layer at the sender's side transforms the data according to the function it handles.
- For this it attaches headers to the data. At the receiver's side, the corresponding layer applies the inverse of the transformation that has been applied at the source.
- As an example, if the Data link layer at the sender's side adds an error detection code to the frame, then at the receiver's side, the Data link layer verifies the error detection code and removes it from the frame before passing it to the next higher level, i.e. the Network layer.

Centralized System

- In Centralized system, the network resources are placed and managed from a main location.
- Centralized network model allows administrators to manage the resources centrally, which other users can access.
- The network servers and other critical network resources are located in a central location in a secure and dedicated server room.

Centralized System characteristics

- One component with non-autonomous parts
- Components shared by users all the time
- All resources accessible
- Software runs in a single process
- Single point of control
- Single point of failure

Distributed System

- A distributed system is a collection of independent computers, interconnected via a network, capable of collaborating on a task.
- A distributed system consists of multiple autonomous computers that communicate through a computer network. The computers interact with each other in order to achieve a common goal.
- A computer program that runs in a distributed system is called a distributed program, and distributed programming is the process of writing such programs.

Distributed System characteristics

- Multiple autonomous components
- Components are not shared by all users
- Resources may not be accessible
- Software runs in concurrent processes
- Different processors
- Multiple points of control
- Multiple points of failure